
Beneath the Mesa

Assessing Aquifer, Well, and Acequia Water Resources Across Sandoval County, New Mexico

A County-Scale Screening Study with Cuba, New Mexico as a Focus Area

Project Proposal

Course: ENG 220

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Project Abstract

Water scarcity is a persistent challenge across New Mexico, and Sandoval County spans several hydrogeologic settings that support communities through groundwater wells, surface water, and acequia irrigation systems. This project will evaluate the spatial and long-term viability of these water resources at a county scale while using Cuba, New Mexico, on the eastern margin of the San Juan Basin, as a focused case study. The study is motivated by concern about declining groundwater availability near Cuba, but rather than assume the severity or cause of that decline, the project will test what available data show about water levels, well characteristics, aquifer properties, water quality, and nearby surface-water and acequia resources. We will integrate complementary public datasets from the U.S. Geological Survey, New Mexico Office of the State Engineer/Interstate Stream Commission, New Mexico Bureau of Geology and Mineral Resources, New Mexico Water Data, and other documented sources.

Analysis will compare aquifer geometry and hydrogeology, historical well and water-level records, groundwater quality and salinity, and the location and role of acequias or community ditches where suitable data are available. Because Sandoval County crosses multiple geologic regions, the county-wide analysis will be used as a screening framework rather than treating the county as one uniform aquifer system. Results will identify areas of relative groundwater stress, data gaps, and potential relationships among wells, aquifers, and surface-water irrigation systems. The findings will be presented in an interactive Streamlit application, with reproducible code and documentation maintained on GitHub and summarized in a final report and seven-minute team video. The final product is intended to provide an accessible, data-driven view of water-resource conditions in Sandoval County and a more detailed assessment of the Cuba area.

Research Points from the Abstract

- **Geographic scope:** Sandoval County, with Cuba as the primary case study rather than the only study area.
- **Groundwater:** compare aquifer geometry, hydrogeology, well characteristics, and historical water-level trends.
- **Water quality:** examine salinity and other available indicators of whether groundwater is suitable for use.
- **Acequias:** map and characterize community irrigation ditches where data are available and examine their relationship to surface-water and groundwater conditions.
- **County-scale comparison:** account for the fact that Sandoval County crosses multiple geologic and hydrogeologic regions.
- **Cuba focus:** test the reported concern of severe aquifer decline using available measurements instead of assuming the conclusion.
- **Data gaps:** identify locations or variables for which monitoring is too sparse to support a strong conclusion.
- **Deliverables:** interactive Streamlit dashboard, GitHub repository, final report, and seven-minute team presentation video.

Proposed Scope and Approach

Expanding the project from Cuba alone to all of Sandoval County is reasonable if the project is structured in two levels. Level 1 is a county-wide screening analysis that maps aquifers, wells, water levels, water quality, and acequia or surface-water features. Level 2 is a deeper Cuba-area case study using the San Juan Basin datasets already identified by the team. This avoids the unrealistic assumption that one aquifer represents the entire county while still responding to the course emphasis on the aquifers, wells, and acequias that support Sandoval County.

The analysis should emphasize measurable indicators rather than attempting a full hydrologic model. Useful indicators include depth to water, change in water level through time, well depth and completion information, aquifer unit, aquifer thickness where available, water-quality or salinity measurements, and the spatial relationship between communities, wells, surface-water features, and acequias. Where records are incomplete, the dashboard should show the limitation explicitly rather than interpolate beyond what the data can support.

Research Questions

1. How do groundwater levels vary across Sandoval County, and which monitored areas show sustained decline, stability, or recovery over the available period of record?
2. What do available well records indicate about well depth, depth to water, and the aquifer units used by communities in different parts of Sandoval County?
3. What do the available data show about groundwater conditions in and around Cuba, and do they support the concern that its principal groundwater source is approaching severe depletion?
4. How do the Ojo Alamo Sandstone and San Jose Formation compare near Cuba in aquifer extent, thickness, hydraulic properties, water levels, and available water-quality measurements?
5. How does groundwater quality, including salinity where available, vary spatially across the county and near Cuba?
6. Where are acequias or community ditches located within the study area, what surface-water sources do they depend on where that information is documented, and which communities or agricultural areas do they serve?
7. Where data permit, do changes in groundwater levels correspond spatially or temporally with nearby surface-water or acequia conditions, or are the available records insufficient to establish a relationship?
8. Which parts of Sandoval County have the strongest monitoring coverage, and where are the most important data gaps for wells, aquifers, water quality, or acequias?
9. Which indicators can be combined into a transparent dashboard to communicate relative water-resource stress without overstating the certainty of the underlying data?

Datasets and Data Acquisition Plan

Preliminary research showed that town-specific data for Cuba are limited and therefore relies on broader datasets that can be filtered geographically. This proposal will retain that approach but organize acquisition around the research questions.

Aquifer Geometry and Hydrogeology

Use USGS digital aquifer and groundwater-model datasets for the San Juan Basin to characterize the Ojo Alamo Sandstone and San Jose Formation near Cuba. For the wider county, use published hydrogeologic mapping and aquifer assessments to distinguish the major geologic regions rather than extrapolating San Juan Basin properties across Sandoval County. Key variables include aquifer extent, thickness/elevation where available, hydraulic properties, and the geologic unit associated with wells.

Well Records and Groundwater Levels

Compile well locations, well depths, completion information, depth-to-water measurements, and repeated water-level observations from the New Mexico Office of the State Engineer, National Groundwater Monitoring Network, Healy Collaborative Groundwater Monitoring Network/Weaver, and New Mexico Water Data. Filter records to Sandoval County, then create a focused subset around Cuba. Repeated measurements should be prioritized for trend analysis; single measurements remain useful for spatial context but should not be treated as time trends.

Water Quality and Salinity

Use New Mexico Bureau of Geology and Mineral Resources datasets, including the San Juan Basin Brackish Water Assessment where applicable, plus other documented water-quality records available through New Mexico Water Data or USGS. Compare water-quality indicators spatially and by aquifer when the source data support that distinction.

Acequias and Surface-Water Context

Add acequias/community ditches as a distinct project layer. Acquire authoritative acequia or community-ditch records, locations, service areas, or project information where available from the New Mexico Office of the State Engineer/Interstate Stream Commission and related state or local sources. Where the data permit, connect acequias to their documented surface-water source and nearby communities or agricultural areas. The project should not assume that an acequia is supplied by groundwater; instead, it should use acequias to show how surface-water irrigation and groundwater-dependent communities coexist within the county.

County Context and Study Boundaries

Use a Sandoval County boundary layer and community/place boundaries to organize the dashboard. Because the county spans several hydrogeologic settings, comparisons should be grouped by region or aquifer system where possible. Cuba remains the detailed case study because the

original proposal already identifies relevant San Juan Basin datasets for that area.

Planned Analysis and Dashboard Outputs

- Interactive county map showing wells, available water-level measurements, major aquifer/geologic regions, and acequia or community-ditch features.
- Cuba-area detail view highlighting the Ojo Alamo Sandstone, San Jose Formation, nearby wells, and available water-level and water-quality records.
- Time-series plots for wells with repeated groundwater-level measurements.
- Summary plots or tables comparing well depth, depth to water, aquifer unit, and water-quality indicators by area where sample size is sufficient.
- Acequia layer showing documented ditch locations and associated surface-water or service information when available.
- Data-coverage layer or table identifying where monitoring is dense, sparse, or absent.
- A clearly labeled interpretation section distinguishing measured trends from hypotheses or areas where the evidence is insufficient.

Primary Sources and Starting Points

- **USGS San Juan Basin digital aquifer data (ScienceBase).**
<https://www.sciencebase.gov/catalog/item/5d8e350be4b0c4f70d0cc484>
- **USGS groundwater-flow model report, WRI 95-4187.**
<https://pubs.usgs.gov/publication/wri954187>
- **New Mexico Bureau of Geology — San Juan Basin Brackish Water Assessment.**
<https://geoinfo.nmt.edu/resources/water/projects/bwa/sjb/home.html>
- **New Mexico Office of the State Engineer — water and well resources.**
<https://www.ose.nm.gov/>
- **National Ground-Water Monitoring Network.**
<https://www.usgs.gov/apps/ngwmn/index.jsp>
- **Healy Collaborative Groundwater Monitoring Network.**
<https://geoinfo.nmt.edu/resources/water/cgmh/home.cfm>
- **New Mexico Water Data Catalog.**
<https://catalog.newmexicowaterdata.org/>
- **New Mexico Interstate Stream Commission — Acequias/Community Ditches.**
https://www.ose.nm.gov/Acequias/isc_acequias.php

Scope Guardrails

- Treat Sandoval County as a comparison area containing multiple aquifer systems, not as one homogeneous groundwater basin.
- Use Cuba as the principal case study so the project remains feasible within the course schedule.
- Treat the statement that Cuba's aquifer is "nearly dead" as a concern to investigate, not as a conclusion unless the data support it.
- Include acequias as part of the county's water-resource system, but avoid claiming groundwater-surface-water interaction unless the available data can support that analysis.
- Prefer reproducible, public datasets that can be downloaded, filtered, and documented in GitHub.

Team Member Contributions

Roles can be adjusted by the team as the project develops.

- **Dylan:** Data acquisition and dataset justification.
- **James:** Data cleaning and preprocessing.
- **Micah:** Writing analysis code and producing visualizations.
- **Meckrem:** Managing GitHub workflow and building the Streamlit application.
- **Petar:** Video editing; building the PowerPoint presentation; writing documentation and the final project report; assisting other members as needed.